1.13 Since the given expression is a quadratic function of c with a positive leading coefficient, its graph opens upward. Therefore, we can differenciate the function with respect to c and set the derivative equal to zero to find the value of c at the minimum point. Because the function is quadratic and opens upward, this minimum point is the absolute minimum. That is,

$$\begin{aligned}
\frac{d}{dc} SS(c) &= \frac{d}{dc} \sum (x_i - c)^2 \\
&= \sum_{i=1}^{n} \frac{d}{dc} (x_i - c)^2 \\
&= \sum_{i=1}^{n} 2(x_i - c) \cdot (-1) \\
&= 2 \sum_{i=1}^{n} (c - x_i) \\
&= 2 \sum_{i=1}^{n} c - 2 \sum_{i=1}^{n} x_i \\
&= 2nc - 2n\bar{x} \quad \left(\because \bar{x} = \frac{\sum x_i}{n} \right) \\
&= 2n(c - \bar{x})
\end{aligned}$$

When $c = \bar{x}$, $SS(c)$ has the smallest value

2.48 a) $P(5 \text{ success in a row}) = (0.8)^5 = 0.3277$

b) $P(1 \text{ miss})$

$= P(\text{miss at } 1^{st} \text{ shot and } 5 \text{ succ}) + P(\text{miss at } 2^{nd} \text{ shot and } 5 \text{ succ})$

$+ P(\text{miss at } 3^{rd} \text{ shot and } 5 \text{ succ}) + P(\text{miss at } 4^{th} \text{ shot and } 5 \text{ succ}) + P(\text{miss at } 5^{th})$

$= (0.2)\cdot(0.8)^5 + (0.8)\cdot(0.2)(0.8)^5 + (0.8)^2\cdot(0.2)(0.8)^5$

$+ (0.8)^3(0.2)(0.8)^5 + (0.8)^4(0.2)(0.8)^5$

$= (0.2)(1 + 0.8 + 0.8^2 + 0.8^3 + 0.8^4)(0.8)^5 = 0.2203$

c) $P(\text{more than 1 miss}) = 1 - [P(0 \text{ miss}) + P(1 \text{ miss})]$

$= 1 - (0.3277 + 0.2203)$

$= 0.4520$

d) The probability of missing a shot before 5 shots in a row is:

$P(\text{miss}) = (0.2)(1 + 0.8 + 0.8^2 + 0.8^3 + 0.8^4)$

$= 0.6723.$

Thus, the pmf is:

$P(X = x) = (0.6723)^x \cdot (0.8)^5$

2.62 X is the # of suits in a 5 card hand

a)

Values X	1	2	3	4
probability	0.00198	0.1459	0.5884	0.2637

$P(2) = \binom{4}{1}\binom{13}{4} \times \binom{3}{1}\binom{13}{1} + \binom{4}{1}\binom{13}{3} \times \binom{3}{1}\binom{13}{2} = 379,236$

$379236 / \binom{52}{5} = 0.1459$

$P(3) = \binom{4}{1}\binom{13}{3} \times \binom{3}{2}\binom{13}{1}\binom{13}{1} + \binom{4}{2}\binom{13}{2}\binom{13}{2} \times \binom{2}{1}\binom{13}{1} = 1,529,100$

$1,529,100 / \binom{52}{5} = 0.5884$

$P(4) = \binom{4}{1}\binom{13}{2} \times (13)\binom{13}{1}\binom{13}{1} = 4(78) \times 13^3 = 685,464$

$685,464 / \binom{52}{5} = 0.2637$

b) $E(X) = 1\cdot P(1) + 2P(2) + 3P(3) + 4P(4) = \boxed{3.113856312 = E(X)}$

2.90

$P(X=x) = \frac{e^{-\lambda}(\lambda)^x}{x!}$ ← # of customers

(a) 20 min = $\frac{1}{3}$ hr

$\lambda = 6(\frac{1}{3}) = 2$

$X \sim Pois(2)$

$P(X=0) = \frac{e^{-2}2^0}{0!} = e^{-2}$

= dpois(0,2) ≈ 0.135

(b) $\lambda = 2$,

$P(X=2) = \frac{e^{-2}2^2}{2!} \approx 0.2707$

= dpois(2,2)

(c) i) $P(X>6) = 1 - P(X \le 6)$

$= 1 - ppois(6,6)$

ppois: less than or equal to

$= 0.393697$

ii) $P(X<6) = P(X \le 5)$

$= ppois(5,6)$

0.445679

iii) $P(X=6)$

$= dpois(6,6) = 0.160623$

(d) $\lambda = 24$

$X \sim Pois(24)$

$P(20 \le X \le 30)$

$= P(X \le 30) - P(X \le 19)$

$= ppois(30,24) - ppois(19,24)$

$= 0.723891$

(e) bc customers may not arrive at constant rate, and there are many variables like rush hours. Also people can come as a group or one person can order several menus.

2.96 a) $H_0 : p_{cash} = p_{ticket}$

$H_a : p_{cash} > p_{ticket}$

Let assume H_0 is true. In other words, the respondents' answers are independent of the given condition (losing cash or ticket). Then, if the probabilities of the responses are the same, we can apply the hypergeometric distribution to test this situation. The model has $\underset{(m)}{22}$ "Yes" and $\underset{(n)}{23}$ "No" responses, and randomly selects $\underset{(k)}{22}$ responses of the cash group from the total 45 responses.

Let X be the number of "Yes" responses from the cash group. Then the pmf is:

$$P(X = x) = \frac{{}_{22}C_x \cdot {}_{23}C_{22-x}}{{}_{45}C_{22}} \quad \text{for} \quad x = 0, 1, 2, \cdots, 22$$

The given table states 13 "Yes" responses of the cash group. Thus, the p-value is:

$$\text{p-value} = P(X \geq 13)$$

$$= \sum_{x=13}^{22} \frac{{}_{22}C_x \cdot {}_{23}C_{22-x}}{{}_{45}C_{22}} = 1 - \text{phyper}(12, 22, 23, 22)$$

$$= 0.1490$$

For $\alpha = 0.05$, p-value is greater than α. Thus, we failed to reject H_0. Therefore, there is not enough evidence to conclude that people who lose cash are more likely to buy a ticket than those who lose a ticket.

b) In the same way, there are 147 "Yes" and 130 "No" with 164 responses from the cash group.

$$X \sim \text{Hyper}(147, 130, 164)$$

$$P(X=x) = \frac{{}_{147}C_x \cdot {}_{130}C_{164-x}}{{}_{247}C_{164}}$$

$$\text{p-value} = P(X \geq 103)$$

$$= 1 - P(X < 102) = 1 - \text{phyper}(102, 147, 130, 164)$$

$$\approx 0.000007$$

For $\alpha = 0.05$, p-value $< \alpha$ which means we reject to $H_0 : p_{cash} = p_{ticket}$. Therefore, there is evidence that people who lose cash are more likely to buy a ticket than those who lose a ticket.

c) The second data set provides stronger evidence because it has a larger sample size.

2.104

a) no, K and Q are not independent. If you have K=3, the values of Q cannot be greater than 2, which means the value of Q depends on the value of K, so they are not independent

b) $P(K=2 \mid Q=2)$

3 cards to fill hand, 48 non queen cards

$\binom{48}{3}$ choices of hand

$\binom{4}{2}$ choices of kings

$\binom{44}{1}$ choices of non king or queen card

$$P(K=2 \mid Q=2) = \frac{\binom{4}{2}\binom{44}{1}}{\binom{48}{3}} \approx 0.015$$

3.1

a) $\int_{-2}^{2} k(x-2)(x+2)\,dx = k\int_{-2}^{2}(x-2)(x+2)\,dx$

$= k\int_{-2}^{2}(x^2-4)\,dx = \left[\left(\frac{1}{3}x^3-4x\right)\right]_{-2}^{2} k$

$= \left[\left[\frac{1}{3}(2)^3-4(2)\right]-\left[\frac{1}{3}(-2)^3-4(-2)\right]\right]k$

$= \left[\left[\frac{8}{3}-8\right]-\left[-\frac{8}{3}+8\right]\right]k$

$= \left[\frac{8}{3}+\frac{8}{3}-8-8\right]k = \left[\frac{16}{3}-16\right]k$

$= \left[\frac{16}{3}-\frac{48}{3}\right]k = -\frac{32}{3}k$

$k = \frac{-3}{32}$

b) $P(X \geq 0) = \int_{0}^{\infty} \frac{-3}{32}(x^2-4)\,dx = \frac{-3}{32}\int_{0}^{2}(x^2-4)\,dx = \frac{-3}{32}\left[\frac{1}{3}x^3-4x\right]_{0}^{2}$

$= \frac{-3}{32}\left[\left(\frac{8}{3}-8\right)-(0)\right] = \frac{-3}{32}\left(\frac{8}{3}-\frac{24}{3}\right)$

$= \frac{48}{32}\left(\frac{16}{96}\right) = \frac{1}{2}$

c) $P(X \geq 1) = \int_{1}^{\infty} \frac{-3}{32}(x^2-4)\,dx = \frac{-3}{32}\int_{1}^{2} x^2-4\,dx = \frac{-3}{32}\left[\frac{1}{3}x^3-4x\right]_{1}^{2}$

$= \frac{-3}{32}\left[\left(\frac{8}{3}-8\right)-\left(\frac{1}{3}-4\right)\right] = \frac{-3}{32}\left(\frac{8}{3}-\frac{24}{3}-\frac{1}{3}+\frac{12}{3}\right)$

$= \frac{-3}{32}\left(\frac{-5}{3}\right) = \frac{5}{32}$

d) $P(-1 \leq X \leq 1) = \int_{-1}^{1} \frac{-3}{32}(x^2-4)\,dx = \frac{-3}{32}\left[\frac{1}{3}x^3-4x\right]_{-1}^{1}$

$= \frac{-3}{32}\left(\frac{1}{3}-4+\frac{1}{3}-4\right) = \frac{-3}{32}\left(\frac{2}{3}-\frac{24}{3}\right) = \frac{-3}{128}\left(\frac{-22}{3}\right) = \frac{11}{16}$

...of your favorite sports team winning ... game
...-off opponent ...

3.4 Show that if f and g are pdfs and $\alpha \in [0,1]$, then $\alpha f + (1-\alpha)g$ is also a pdf

Must show that $\alpha f + (1-\alpha)g \geq 0$, and $\int \alpha f + (1-\alpha)g \, dx = 1$

First we know that $f(x) \geq 0$ and $g(x) \geq 0$ (def of pdf).

Then, $\alpha \in [0,1]$, so $\alpha \geq 0$ and $(1-\alpha) \geq 0$.

Therefore $\alpha f(x) \geq 0$ and $(1-\alpha)g(x) \geq 0$ and it follows that $\alpha f(x) + (1-\alpha)g(x) \geq 0$.

Then, we also know $\int_{-\infty}^{\infty} f(x)\,dx = 1$ and $\int_{-\infty}^{\infty} g(x) = 1$.

take $\int_{-\infty}^{\infty} [\alpha f(x) + (1-\alpha)g(x)]\,dx$

$$\Rightarrow \int_{-\infty}^{\infty} \alpha f(x)\,dx + \int_{-\infty}^{\infty} (1-\alpha)g(x)\,dx$$

$$\Rightarrow \alpha \int_{-\infty}^{\infty} f(x)\,dx + (1-\alpha)\int_{-\infty}^{\infty} g(x)\,dx = \alpha + 1 - \alpha = 1$$

$\blacksquare$

3.10 cdf $F(x) = \frac{x^2}{4}$ on $[0,2]$ and $Y = F(X) = \frac{X^2}{4}$

a) $P(X \le 1) = \int_0^1 f_X(x)\,dx = F(1) - F(0) = \frac{1}{4} = 0.25$

b) $P(X \le \frac{1}{4}) = \int_0^{\frac{1}{4}} f(x)\,dx = F(\frac{1}{4}) - F(0) = (\frac{1}{4})^2/4 = \frac{1}{64} = 0.0156$

c) pdf $f(x) = F'(x) = (\frac{x^2}{4})' = \frac{x}{2}$

d) $P(Y \le 1) = P(\frac{X^2}{4} \le 1)$

$= P(X^2 \le 4)$

$= P(0 \le X \le 2)$ $(\because X \in [0,2])$

$= \int_0^2 f(x)\,dx = 1$

e) $P(Y \le \frac{1}{4}) = P(\frac{X^2}{4} \le \frac{1}{4})$

$= P(X^2 \le 1)$

$= P(0 \le X \le 1)$ $(\because X \in [0,2])$

$= \int_0^1 f(x)\,dx = 0.25$

f) $F(Y) = P(Y \le y)$

$= P(\frac{X^2}{4} \le y)$

$= P(0 \le X \le 2\sqrt{y})$

$= F(2\sqrt{y})$

$= \frac{(2\sqrt{y})^2}{4} = y \quad (0 \le y \le 1)$

$$\Rightarrow F(Y) = \begin{cases} 0 & (y<0) \\ y & (0 \le y \le 1) \\ 1 & (y>1) \end{cases}$$

g) $f(y) = F'(Y) = 1 \quad (0 \le y \le 1)$

h) Uniform distribution. That is, $Y \sim Unif(0,1)$